KEUN-SOO HEO (허근수)

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RESEARCH INTERESTS

- **Medical Image Analysis for Medical Imaging**
  - Resting-state fMRI Denoising, Chest X-ray Report Generation
- **Trustworthy Medical AI**
  - Disease-aware Image-Text Alignment, Self-correction, Hallucination Mitigation
- **Data-Efficient Learning under Medical Data Constraints**
  - Domain Adaptation, Meta-Learning, Semi-Supervised Learning

EDUCATION

Korea University — **Seoul, South Korea**
Integrated M.S.-Ph.D. in Department of Artificial Intelligence — Mar. 2020 – Feb. 2027 (Expected)
• Medical Artificial Intelligence Lab. (Advisor : Prof. Tae-Eui Kam) — Jan. 2020 – Present
• GPA: 4.31/4.5

The Catholic University of Korea — **Bucheon, Gyeonggi-do, South Korea**
B.S. in Department of Information, Communication and Electronic Engineering — Mar. 2014 – Feb. 2020
• Image Signal Processing Lab. (Advisor : Prof. Changwoo Lee) — Dec. 2017 – Jan. 2020
• GPA: 4.33/4.5
* Served in the Republic of Korea Army as part of mandatory military duty (2015–2017)

RESEARCH EXPERIENCE

- **Resting-state fMRI Denoising & Domain Adaptation**
  - Meta-learning-based semi-supervised domain adaptation for sparsely labeled fMRI denoising (MICCAI 2025 Oral, Top 2%)
  - Unified spatio-temporal-spectral fusion framework for fMRI noise removal (IEEE JBHI 2024)
  - Deep attentive spatio-temporal feature learning for fMRI denoising (NeuroImage 2022)
- **Multi-modal Medical Vision-Language Models**
  - Aligned image-text representations to reduce hallucinated findings in chest X-ray report generation (CVPR 2025)
  - Summarized multi-view reports via LLM-based ensemble (EMBC 2025, Oral)
  - Anatomy-guided token proposal for 3D CT VQA (SMC 2026)

TEACHING EXPERIENCE

- **AI Training Program for LG CNS**, Korea University
  - Teaching Assistant, Data AI (Instructor: Prof. Sejun Park) Sep. 2023 – Nov. 2023
  - Teaching Assistant, Machine Learning (Instructor: Prof. Tae-Eui Kam) May 2023 – Jun. 2023

SCHOLARSHIPS

- **KOBRA Travel Award (KOBRA-TA)**, Korea Bio Research Association
  - Jun. 2025 — KRW 3,000,000
- **BK21 FOUR Graduate Research Scholarship**, Ministry of Education, Korea
  - Spring 2022, Fall 2021, Spring 2021
- **Graduate School Scholarship,** Korea University
  - Fall 2020
- **Research Assistantship,** Korea University
  - Spring 2021
- **Outstanding Academic Scholarship**, The Catholic University of Korea
  - Fall 2019, Spring 2019, Fall 2018, Spring 2018, Fall 2015
- **Presidential Entrance Scholarship**, The Catholic University of Korea
  - Spring 2014, Fall 2014

AWARDS

- MICCAI 2025 Oral Presentation & Highlight Paper, 2025
- First Place Graduate in Department (Valedictorian), The Catholic University of Korea, 2020
- Honorable Mention Award, Seoul App Development Competition, Seoul Metropolitan Government, 2019

SELECTED PUBLICATIONS

Keun-Soo Heo, Ji-Wung Han, Soyeon Bak, Minjoo Lim, Bogyeong Kang, Sang-Jun Park, Weili Lin, Han Zhang, Dinggang Shen, and Tae-Eui Kam, "Sparsely Labeled fMRI Data Denoising with Meta-Learning-Based Semi-Supervised Domain Adaptation," *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025. (Oral Presentation (Top 2%), Early Accepted (Top 9%))

Sang-Jun Park*, **Keun-Soo Heo***, Dong-Hee Shin, Young-Han Son, Ji-Hye Oh, and Tae-Eui Kam, "DART: Disease-aware Image-Text Alignment and Self-correcting Re-alignment for Trustworthy Radiology Report Generation," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. (*: Equal contribution)

Keun-Soo Heo, Dong-Hee Shin, Sheng-Che Hung, Weili Lin, Han Zhang, Dinggang Shen, and Tae-Eui Kam, "Deep Attentive Spatio-Temporal Feature Learning for Automatic Resting-State fMRI Denoising," *NeuroImage*, 2022. (JCR IF: 5.7, Top 7.14% in NEUROIMAGING)

PUBLICATIONS

Yubin Jeong, **Keun-Soo Heo**, and Tae-Eui Kam, "Anatomy-Guided Abnormality Token Proposal for 3D CT Visual Question Answering," *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2026

Minjoo Lim, Bogyeong Kang, Hyun Jung Lee, Eunjung Jo, **Keun-Soo Heo**, Mingxia Liu, and Tae-Eui Kam, "Retrieval-based Mixture-of-Experts for Patient-Specific Cancer Survival Prediction with Incomplete Multimodal Data," *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026.

Bogyeong Kang, Minjoo Lim, Hyeonyeong Nam, **Keun-Soo Heo**, Mingxia Liu, and Tae-Eui Kam, "Tumor-Aware Direct Preference Optimization for Longitudinal Post-Operative MRI Generation," *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026.

Jun-Mo Kim*, Woohyeok Choi*, Sang-Jun Park, **Keun-Soo Heo**, Young-Han Son, Ji-Hye Oh, Dong-Hee Shin, and Tae-Eui Kam, "SeeEEG: Semantic-aware EEG-based Multi-Modal Retrieval-Augmented Generation for High-Fidelity Visual Brain Decoding," *International Conference on Computer Vision (ICCV) Workshop HiCV*, 2025.

Bogyeong Kang, Sang-Jun Park, Minjoo Lim, Myeongkyun Kang, **Keun-Soo Heo**, Ji-Hye Oh, Hyun Jung Lee, and Tae-Eui Kam, "Pre-to-Post Operative MRI Generation with Retrieval based Visual In-Context Learning," *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI),* 2025.

Sang-Jun Park, **Keun-Soo Heo**, Bogyeong Kang, Minjoo Lim, and Tae-Eui Kam, "Group-wise Compression and Summarization via LLM-based Ensemble for Chest X-ray Report Generation," *International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2025. (**Oral Presentation**)

Bogyeong Kang, Hyeonyeong Nam, Myeongkyun Kang, **Keun-Soo Heo**, Minjoo Lim, Ji-Hye Oh, and Tae-Eui Kam, "Target-Aware Cross-Modality Unsupervised Domain Adaptation for Vestibular Schwannoma and Cochlea Segmentation," *Scientific Reports*, 2024.

Jun-Mo Kim, **Keun-Soo Heo**, Dong-Hee Shin, Hyeonyeong Nam, Dong-Ok Won, Ji-Hoon Jeong, and Tae-Eui Kam, "A Learnable Continuous Wavelet-based Multi-Branch Attentive Convolutional Neural Network for Spatio-Spectral-Temporal EEG Signal Decoding," *Expert Systems With Applications*, 2024.

Sanghyeon Cho, Bogyeong Kang, **Keun-Soo Heo**, Eunjung Jo, and Tae-Eui Kam, "Enhanced Structure Preservation and Multi-View Approach in Unsupervised Domain Adaptation for Optic Disc and Cup Segmentation," *IEEE International Symposium on Biomedical Imaging (ISBI),* 2024.

Minjoo Lim, **Keun-Soo Heo**, Jun-mo Kim, Bogyeong Kang, Weili Lin, Han Zhang, Dinggang Shen, and Tae-Eui Kam, "A Unified Multi-Modality Fusion Framework for Deep Spatio-Temporal-Spectral Feature Learning in Resting-State fMRI Denoising," *IEEE Journal of Biomedical and Health Informatics*, 2024.

Bogyeong Kang, Hyeonyeong Nam, Ji-Wung Han, **Keun-Soo Heo**, and Tae-Eui Kam, "Multi-view Cross-Modality MR Image Translation for Vestibular Schwannoma and Cochlea Segmentation," *MICCAI BrainLes Workshop*, 2022.

TECHNICAL SKILLS

- Programming Languages: Python, C++, MATLAB
- Programming Frameworks: PyTorch, TensorFlow
- Medical Imaging Tools: FSL, SPM, FreeSurfer, DICOM
- Domains: fMRI, Chest X-ray, Radiology Report